

Optimised Route Allocation System

Computer Science GCE A Level Non-Examined Assessment

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September 1, 2026

Project Summary

This project will design, develop, and evaluate an optimised route allocation system for an organisation such as a supermarket, courier business, or local delivery company. The system will help allocate deliveries to multiple drivers while attempting to minimise total travel distance, reduce inefficient routing, and respect real-world constraints such as vehicle capacity, route length, customer delivery time windows with respect to tolls.

The underlying optimisation problem is the Capacitated Vehicle Routing Problem with Time Windows, abbreviated CVRPTW. In this well-known computer science problem, a fleet of vehicles must visit a set of customer locations. Each vehicle has limited capacity, and each customer may only be served within a specified time period. Although finding a perfectly optimal solution can be computationally expensive for larger inputs, the project will aim to produce high-quality routes using algorithms and heuristics that are practical for a real delivery environment.

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1 Introduction

Many supermarkets and delivery-based companies must organise large numbers of daily deliveries. Each order has a destination, a demand value (such as weight or number of items), and often a delivery window. At the same time, each delivery vehicle has a maximum carrying capacity and each driver may only be able to travel for a limited distance or duration. If routes are planned manually, the process can be slow, inconsistent, and inefficient as such routes may lead to unnecessary fuel costs, late deliveries, underused vehicles, and increased pressure on staff (which may result in accidents).

The aim of this project is to create an optimised route allocation system that supports a delivery manager in assigning deliveries to drivers. The system will take a set of delivery requests and vehicle details as input, then generate routes that attempt to satisfy all constraints while keeping the overall route cost as low as possible. Route cost may include total distance, number of vehicles used, estimated travel time, or a combination of these factors. The final system should present the generated routes clearly so that a user can understand which deliveries have been allocated to each driver and in what order they should be completed.

This project is suitable for a Computer Science GCE A Level Non-Examined Assessment because it involves several important computational concepts. These include graph representation, data modelling, algorithm design, optimisation, validation, testing, and evaluation. Locations can be represented as nodes in a graph, while travel distances or times can be represented as weighted edges. The system can then apply a route construction method, such as a nearest-neighbour or savings-based heuristic, followed by improvement techniques to reduce inefficient journeys. Constraint checking will also be required to ensure that no route exceeds vehicle capacity, route length, or delivery time window limits.

The intended user is a delivery coordinator working for a supermarket or company that manages multiple delivery drivers. The user needs a system that is faster and more reliable than planning routes manually. The system should allow the user to enter, load, or edit delivery and vehicle data, generate optimised route allocations, and view the results in a clear format. If a delivery cannot be allocated because of the given constraints, the system should identify this clearly rather than producing an invalid route.

The project will focus on producing a functional and testable prototype rather than a commercial-scale logistics platform. The prototype will use a realistic sample dataset containing customer locations, demands, vehicle capacities, time windows, and route limits. The success of the project will be judged by whether the system can create valid routes, reduce total travel time compared with a simple baseline approach, and provide understandable outputs for the user.

1.1 Problem Statement

A delivery company needs to assign a set of deliveries to a limited number of vehicles. Each delivery has a location, demand, and delivery time window. Each vehicle has a maximum capacity and may have a maximum route length. The system must decide which deliveries should be assigned to each

vehicle and the order in which they should be visited, while aiming to minimise route cost and avoid breaking any constraints.

1.2 Project Aim

The aim of this project is to develop an optimised route allocation system that can generate efficient and valid delivery routes for multiple drivers while respecting vehicle capacity, delivery time windows, and maximum route length constraints.

1.3 Initial Objectives

- Analyse the needs of a supermarket or delivery company that manages multiple vehicles and customer deliveries.
- Model the delivery network using suitable data structures, including customer nodes, vehicle records, demands, time windows, and distance or travel-time values.
- Design and implement an algorithm that constructs valid vehicle routes while checking capacity, time window, and route length constraints.
- Improve the generated routes using an optimisation technique or heuristic so that the system performs better than a basic baseline method.
- Provide a clear user interface or output format showing each driver's assigned route, delivery order, total load, and route distance or duration.
- Test the system using normal, boundary, and erroneous data to confirm that constraints are handled correctly.
- Evaluate the final system against the project aim, user requirements, and measurable performance criteria.

2 Analysis

2.1 Background

Over the past few decades, the world has bared witness to a surge in globalisation, which has given society a greater access to products from around the world, faster communication, new opportunities, and more convenient ways of living. With just a few clicks, we can now order almost anything and have it delivered to our doorstep. Specifically, Dubai and the wider UAE have seen an exponential growth in deliveries of parcels including food, groceries and e-commerce. Platforms such as Talabat, Noon, Careem, Deliveroo and newer entrants (Keeta) have driven an expansion of last-mile logistics, with estimates stating tens of thousands of workers are working in the sector nationwide, and around 80,000+ working across logistics and delivery ecosystem as a whole. This growth can be accredited towards the high smartphone penetration, on-demand consumer expectation and the post-pandemic normalisation of online ordering. Tax Free Salaries 2026

Despite this demand, delivery drivers occupy the lower end of the UAE wage distribution. Aggregated salary data places typical delivery driver earnings in the range of about AED 2,500-4,500 per month, with many roles clustering around AED 3,000-3,500. In practice, riders often work 10-12 hour shifts, six days a week, to achieve these figures, either on fixed salaries with performance bonuses or on a commission-only model where they are paid per completed order. Commission rates reported in the UAE commonly fall between roughly AED 7.5 and AED 10 per trip, aligning with riders' own accounts of "7-9 dirhams an order," and requiring roughly 15-20 deliveries per day merely to break even after covering fuel, food, accommodation, and visa or agency fees. Indeed 2026

This creates a downstream risk which creates systemic penalties and safety vulnerabilities. There is a transfer of financial and operational risk onto the driver, as compensation is performance-oriented or proportional to order volume, time becomes a rider's currency. Any unexpected delays (whether from traffic congestion, restaurant preparation latency or gated community security checks) directly decrease their hourly wage and risk a deficit by pushing them below a financial break-even threshold needed to cover operational costs - fuel, bike leases and living costs. This economic structure creates a high pressure environment and feedback loop that directly incentivises high risk behaviours on the road. RoadSafetyUAE 2024

1. Algorithmic pressure and customer ratings. Delivery algorithms track live trip time, punishing slow drivers with lower dispatch priorities or reduced access to peak hour shifts. Moreover, customer dissatisfaction over late orders leads to poor ratings which in turn can trigger platform suspensions. The Jerusalem Post and The Media Line 2024
2. Compounding financial penalties. When delays are financially threatening, drivers absorb the costs. To make up for lost time, drivers often frequently engage in reckless driving practices such as lane weaving, driving in fast lanes and speeding through pedestrian zones. Lane Bans and accident Stats (854 in 2024, 962 in 2025) Dubai Police and Roads and Transport Authority 2025
3. Rising casualty rates and health impacts. The combination of extreme heat (in the Gulf Regions), 10-12+ hour shifts and high speeds leads to extreme fatigue, resulting in more

frequent road accidents. Emergency medicine data reports that riders disproportionately suffer severe orthopedic and traumatic brain injuries due to high speed collisions. The Media Line 2025

2.2 The Client

As on-demand deliveries has become a very large operational pillar for local businesses, there is a widespread need for intelligent route optimisation. Implementing a Capacited Vehicle Routing solution directly addresses this demand by streamlining dispatch schedules and lowering vehicle running costs. There are many who may benefit from this solution, ranging from local cloud kitchens to independent courier fleets. Such solution can maximize operational efficiency and profit margins while actively improving daily working conditions for drivers by reducing peak-hour stress and unnecessary travel. Potential clients include:

Local businesses:

1. Gourmet catering or meal-prep. Dubai has a huge meal-prep delivery ecosystem. These businesses operate by utilising small driver fleets facing strict time windows (often during lunch break in corporate offices) across a very wide span of area -JLT, Dubai Marina or Downtown.
2. Boutique florists or specialty bakery. Companies delivering custom cakes, party items or flowers, face strict capacity constraints (e.g delicate and fragile goods per vehicle) and have a tight delivery schedule.
3. Independent laundry and dry cleaning services. Operations offering scheduled home pickups and drop-offs operate on recurring and different routes.
4. Local cloud kitchen or direct-delivery restaurant. Small independent kitchens using 3 to 5 in-house drivers for direct phone/website orders (rather than aggregator fleets) need route planning for clustered orders.
5. School bus departure from school. Large schools often utilise private transport operators (such as STS - School Transport Services), where they must drop off students at a variety of different locations.

Logistics and fleets:

1. Local courier fleet. A dispatcher or operations coordinator at a small scale logistics agency in areas such as Al Qouz, Diera or IMPZ.
2. E-commerce fulfillment. Any e-commerce or online store running in Dubai who dispatches orders via inhouse drivers.

Personal proxies

1. Connections in related fields. A parent, relative or family friend who works and manages scheduling, deliveries or logistics in their main field of work.

2.2.1 Evaluation of Potential Stakeholders

I have formatted my options for the intended client into a table.

Prospective Client / Role	Operational Fit (CVRPTW)	Accessibility & Dialogue	Verdict / Decision
Option A: Small Local Courier Fleet Supervisor	High (Manages multiple vehicles, daily drop-offs, routes vary).	Moderate (Requires corporate permissions/scheduling).	Secondary Option: Good backup, but harder to guarantee rapid feedback loops for testing.
Option B: Daily Meal-Prep Catering Dispatcher	High (Strict morning time windows, fixed vehicle capacities, localized Dubai zones).	High (Direct access to manager/owner).	SELECTED: Perfect balance of real-world constraints (time windows + capacity) and user availability.
Option C: Relative / Family Friend in Logistics (Proxy)	Moderate to High (Understands operational pain points).	Very High (Immediate availability for interviews & evaluation).	Alternative Proxy: Reliable fallback if business access falls through.

Table 1: Evaluation of Potential Stakeholders

Option B (meal-prep delivery) was selected as the primary client as unlike large logistic firms, where route planning and dispatch algorithms are executed by large corporate industrial software, this business relies on manual scheduling via excel sheets

2.3 Analysis of Existing Systems

Commercial-grade route planning applications are designed to support more than navigation between two locations. A delivery coordinator can provide a collection of orders, customer addresses, delivery time windows, vehicle capacities and driver availability. The software can then allocate stops between vehicles, determine an efficient visiting order and distribute the resulting routes to drivers. Many delivery-management platforms also provide live driver tracking, estimated arrival times, customer notifications and proof-of-delivery records, allowing the coordinator to monitor orders from dispatch through to completion. Routific 2026; Route4Me 2026

The available systems can be divided into different categories. Google OR-Tools is a developer-focused optimisation library capable of modelling vehicle capacities and customer time windows, but it does not provide a complete interface for a delivery coordinator. Google for Developers 2026b In contrast, subscription platforms such as Routific and Route4Me combine route optimisation with

dispatcher dashboards and mobile tools for drivers. Comparing these systems with the client’s current spreadsheet-based process will identify useful features, limitations and opportunities for the proposed solution. The comparison will focus on suitability for a small meal-prep business rather than attempting to reproduce every feature of a large commercial logistics platform.

2.3.1 Routific Application Navigation

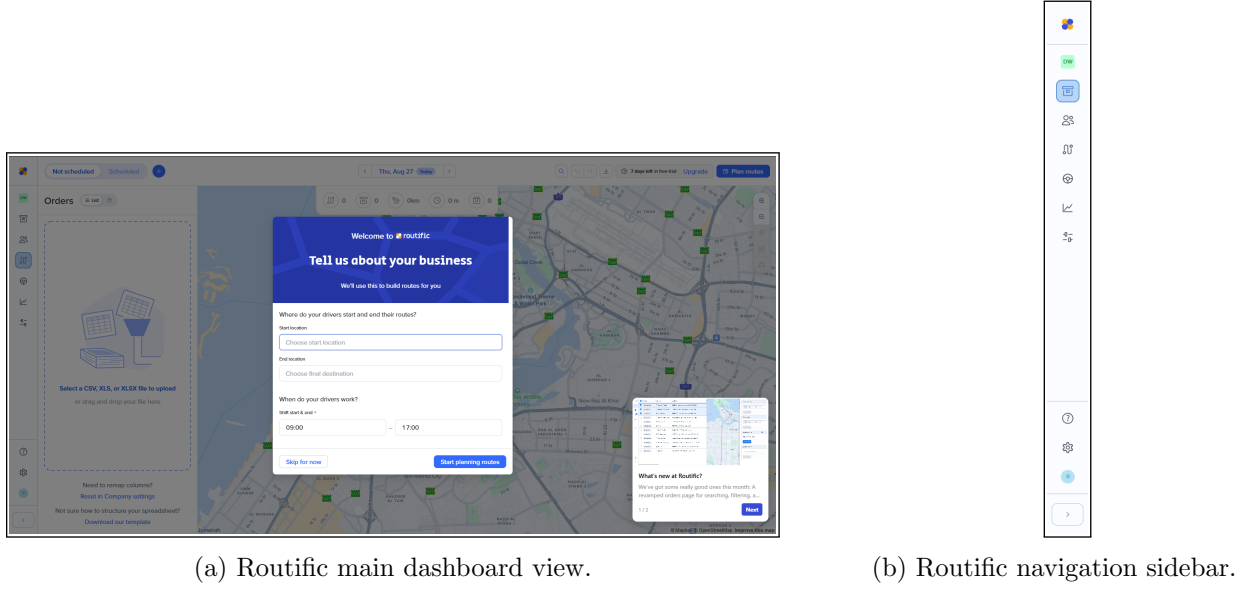
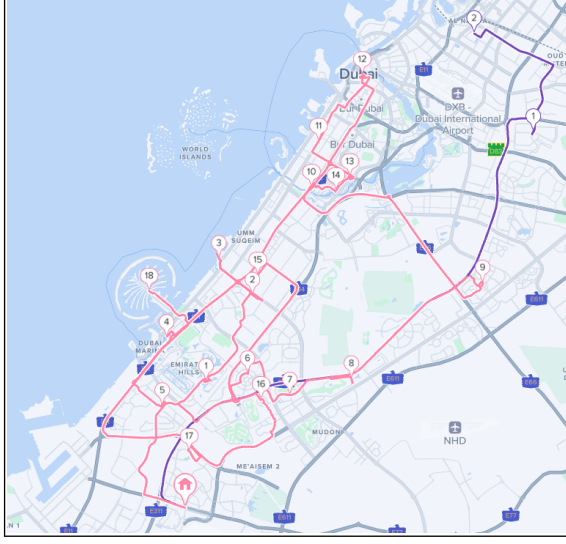


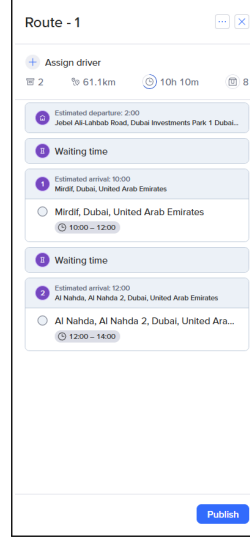
Figure 1: Routific capacity and time window interface.

Figure 1a shows the full-screen dashboard displayed after the user logs in to Routific. The navigation sidebar appears on the left, with the Orders button highlighted to indicate that the order-management workspace is active. A map of Dubai (using OpenStreetMap) is displayed in the background, while input fields in the foreground allow the user to enter each driver’s starting location and shift start time.

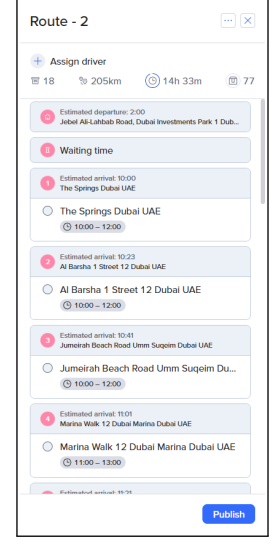
Figure 1b shows the navigation sidebar displayed when the Routific route-planning interface is opened. It contains the following options, listed from top to bottom: (a) workspaces, (b) orders (c) customers (d) routes (e) drivers (f) insights (g) route prefrencing (h) help (i) company settings (j) user settings and log out



(a) Overall optimised routes map.



(b) Optimised Route 1.



(c) Optimised Route 2.

Figure 2: Routific’s overall optimised route map and the two individual driver routes.

To evaluate how commercial solvers handle the constraints identified in Section 1.3 for CVRPTW a test dataset of 20 localized delivery orders was generated and imported into Routific (by using Comma-Separated Values format). Each record and value was configured with a specific geographic location, a volumetric load value (metric based from 0-10, calculated by mathematical $\sum(\text{Load of Assigned Orders}) \leq \text{Vehicle Capacity Limit}$) and a strict delivery time window to stress-test the system’s scheduling algorithm. A small snippet of the sample data set is provided here.

Order ID	Address / Coordinates	Time Window	Load
ORD-001	Dubai Marina, Tower A	08:00 - 10:00	4
ORD-002	Jumeirah Lakes Towers, Cluster C	08:30 - 11:00	2
ORD-003	Business Bay, Executive Towers	10:00 - 12:30	5
ORD-004	Downtown Dubai, Boulevard	14:00 - 16:00	3

Table 2: Sample extract of the custom CSV dataset imported into Routific (see Appendix A for the full dataset).

Figure 2a together showcases the algorithm’s result of the automatic ”Schedule all” optimisation. Both Figure 2b and Figure 2c contain the following headlined metrics:

Metric	Overall	Optimised Route 1	Optimised Route 2
Routes	2	1	1
Orders	20	2	18
Distance	266.07 km	61.1 km	205 km
Working time	24 h 43 min / 31 h	10 h 10 min / 15 h 30 min	14 h 10 min / 15 h 30 min
Capacity	85	8	77

Table 3: Summary of Routific’s overall optimisation and individual route metrics.

2.3.2 Findings

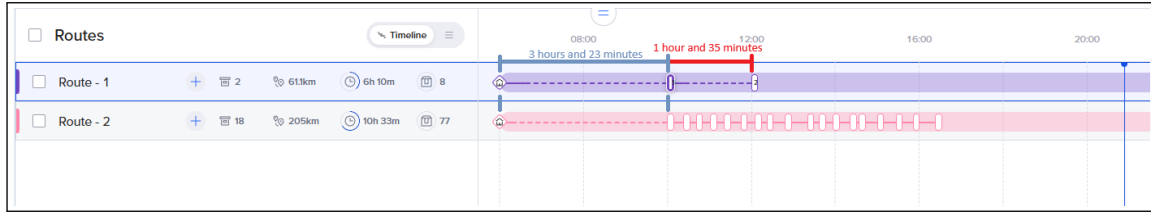
The solution generated by Routific’s inbuilt algorithm unveiled a substantial imbalance between the two drivers. Optimised route 1 (Figure 2b) was assigned only 2 of the 20 orders, a combined load of 8 and a distance of 61.1 kilometres. In stark contrast, Optimised route 2 recieved the remaining 18 orders, a combined load of 77 and approximately 205 kilometres. Although every order was allocated and two routes were made, the workload was not distributed evenly as Optimised route two contained 90% of the orders and around 91% of the total load. This imbalance is notable given the driver welfare concerns raised in Section 2.1. An optimised route allocation system that minimises total distance without workload balance risks reproducing the same over-burdening dynamics aforementioned by concentrating the pressure onto the drivers.

Furthermore, Routific also produced a solution using only two drivers, although the dataset only contained order information and did not itself establish a set amount of drivers or that 2 drivers would be an appropriate fleet size. This suggests that the number of available drivers in Optimised route map (Figure 2a) may have been inferred by the route-planning algorithm rather than calculated from the imported orders. Consequently, a system designed for the intended client should require the fleet size, vehicle capacities and driver shift limits to be entered explicitly rather than anticipation or estimation. To determine whether this result was a of a single occurrence, I submitted the same dataset again again. Routific reproduced the exact same two layouts and the same uneven distributions. I also altered the drivers’ shift times but the imbalance remained. Reproducing the results under an altered conditions provides evidence that the outcome was not an isolated anomaly but may instead be caused by Routific’s internal processes.

The difference in working time was also of concern. One driver was given a comparatively small number of deliveries (2) and a long period without any deliveries being made, yet was allocated approximately 9 hours of working time. As an independent comparison, I entered the locations assigned to Optimised Route 2 into Google Maps. At the time of capture and under low-traffic conditions, Google Maps estimated a total driving time of approximately 53 minutes, as shown in Figure 3b. This estimate is not directly equivalent to Routific’s working-time figure because the systems use different path-finding methods and Google Maps does not include service durations, waiting for time windows or breaks. Nevertheless, the large difference suggests that a significant proportion of the scheduled time resulted from waiting and scheduling constraints rather than transportation alone. The break periods allocated by Routific are shown in Figure 3a.

Another potential source of error is geographic ambiguity. Different streets, buildings or districts may share the same or a very similar name, meaning that an incomplete address could be matched to

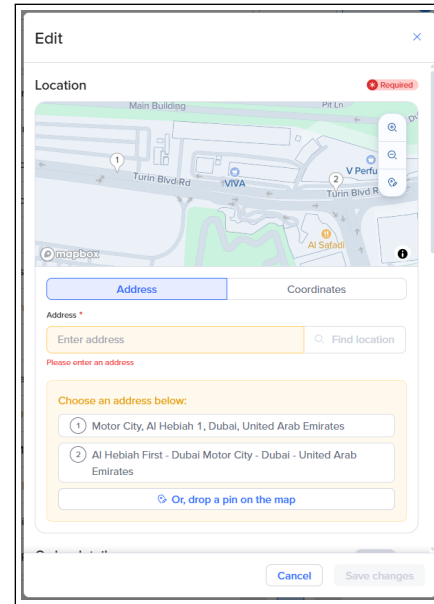
the wrong coordinates. As demonstrated in Figure 3c, this could place a delivery in an unintended location and consequently distort the calculated distance, journey time and allocation of orders between drivers. To reduce this risk, my system should geocode and validate every address before route planning begins. Where multiple matches are found, it should display the standardised address and map location for user confirmation, request additional information such as an area name or coordinates, and prevent optimisation until the ambiguity has been resolved.



(a) Routific schedules for both routes, including the allocated break periods.



(b) Independent Google Maps estimate of approximately 53 minutes under low-traffic conditions.



(c) Geographic mix-up identified in Routific's generated route allocation.

Figure 3: Comparison of Routific's generated schedules with independent driving-time and geographic evidence.

2.3.3 Summary of Routific Analysis

The investigation identified several design choices that may make Routific unsuitable for the intended small-business client and that should be addressed differently in my own system:

1. Workload imbalance. Routific produced routes containing 2 and 18 stops respectively, with no visible mechanism for balancing work between drivers. My optimisation method should

therefore penalise large differences in stop count and allocated load, rather than considering only the total distance or overall fleet load.

2. Capacity applied after route construction. In the observed workflow, routes were clustered before a specific driver and vehicle capacity were attached. This creates uncertainty over whether an existing route will be rejected or rebuilt when the assigned vehicle has insufficient capacity. My system should check vehicle capacity while constructing and improving each route, rather than treating it as a downstream assignment problem.
3. Manual geographic resolution. Ambiguous addresses required the user to identify and correct the selected map location manually. My system should detect multiple geocoding matches before optimisation and require the user to confirm the intended address or coordinates.
4. Subscription access. The interface displayed a limited free trial and an “Upgrade” prompt, indicating that continued use requires a paid subscription. This may be a barrier for a small meal-prep business, whereas the proposed bespoke system would not impose a recurring software-subscription charge on the client.

These findings show that an effective route is not defined by distance alone. For the intended client, route validity, balanced driver workloads, reliable address validation and affordable access must also be treated as important system requirements for end-user satisfaction.

2.4 Evaluation of Pre-existing Systems

Having conducted an in-depth empirical evaluation of Routific in Section 2.3, this section situates those findings alongside the other systems considered in this project’s existing-systems research. It is important to distinguish between the depth of evidence available for each system: the findings on Routific are based on direct hands-on testing, including a custom dataset, repeated trials and cross-verification against independent data. Route4Me and Google OR-Tools were not tested directly, so their entries below reflect capabilities stated in their providers’ official documentation. Google for Developers 2026a; Google for Developers 2024; Route4Me Support 2026a; Route4Me Support 2026b

System	Approach	Handles capacity/time windows?	Cost/access	Relevant strength	Relevant limitation (for your user)
OR-Tools	Constraint programming and metaheuristics	Yes: capacity constraints and time windows are configurable-Google for Developers 2026a; Google for Developers 2024	Free, but requires developer integration	Industry-strength solver	Not directly usable by a non-technical coordinator; it is a library rather than a complete product
Routific	SaaS	Capacity was applied after clustering in the observed workflow (Section 2.3.3); time windows produced unexplained idle time (Figure 3a)	Subscription	Polished interface and dispatch features	Ongoing cost and the workload imbalance identified during direct testing
Route4Me	SaaS	Provider documentation states that capacity and time-window constraints are supported-Route4Me Support 2026a; Route4Me Support 2026b	Commercial access	Supports configurable operational constraints	Not tested directly; some constraint features must be enabled separately
Manual planning (spreadsheet or WhatsApp)	N/A	No automated checking or built-in route optimiser.	Free	Familiar and requires no setup	Error-prone, does not scale and matches the interview’s reported pain points

2.5 Client Interview

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Appendix A Routific Sample Data

This appendix contains the unedited sample data imported into Routific’s automated route allocation system. The resulting routes are shown in Figure 2.

Table 4: Routific Sample Order Data

Order	Address	Start	End	Load	Duration (mins)	Notes
Order 1	Marina Walk 12, Dubai Marina, Dubai, UAE	11:00	13:00	4	5	Vegetarian meal boxes
Order 2	Al Wasl Road 45, Jumeirah 1, Dubai, UAE	12:00	14:00	6	5	Leave with concierge
Order 3	Sheikh Zayed Road, Business Bay, Dubai, UAE	11:30	13:30	3	5	Office delivery, ring bell
Order 4	Al Barsha 1, Street 12, Dubai, UAE	10:00	12:00	5	5	Gate code 4521
Order 5	Downtown Boulevard, Dubai, UAE	13:00	15:00	2	5	Call on arrival
Order 6	Jumeirah Beach Road, Umm Suqeim, Dubai, UAE	10:00	12:00	4	5	No pork items
Order 7	Al Quoz Industrial 3, Dubai, UAE	14:00	16:00	8	10	Bulk order for staff canteen
Order 8	Motor City, Dubai, UAE	11:00	13:00	3	5	Fragile packaging
Order 9	Dubai Silicon Oasis, Dubai, UAE	12:30	14:30	4	5	Leave at reception
Order 10	Arabian Ranches, Dubai, UAE	10:30	12:30	5	5	Gluten free meals
Order 11	Al Furjan, Dubai, UAE	13:00	15:00	3	5	Call before arrival
Order 12	Dubai Sports City, Dubai, UAE	14:30	16:30	6	5	Weekly subscription box
Order 13	Jumeirah Village Circle, Dubai, UAE	11:00	13:00	2	5	High protein plan
Order 14	Palm Jumeirah, Dubai, UAE	15:00	17:00	4	10	Villa delivery, gate access needed
Order 15	Mirdif, Dubai, UAE	10:00	12:00	3	5	Standard plan
Order 16	Al Nahda, Dubai, UAE	12:00	14:00	5	5	Family size order
Order 17	Deira Creek, Dubai, UAE	13:30	15:30	4	5	Office building, 3rd floor
Order 18	Dubai Investment Park, Dubai, UAE	14:00	16:00	7	10	Corporate catering order
Order 19	Discovery Gardens, Dubai, UAE	11:00	13:00	3	5	Vegan plan
Order 20	The Springs, Dubai, UAE	10:00	12:00	4	5	Standard plan